Assume/Guarantee Contract of `ModeSwitch`

May 21, 2026

1 Variable Declaration

Maps : `variables` in Code $\mapsto$ variables in Formal Specification

Alphabet :
pGyroRate, flgSP, royaw, piyaw, m_workMode, m_workMode′, m_countMode, m_countMode′, m_countPublic, m_countPublic′, time_D2P, time_D2P_overtime, clock, clock′

Inputs :

- `pGyroRate` $\mapsto$ pGyroRate : $\mathbb{R}^*$
- `flgSP` $\mapsto$ flgSP : $\mathbb{R}$
- `royaw` $\mapsto$ royaw : $\mathbb{R}$
- `piyaw` $\mapsto$ piyaw : $\mathbb{R}$

Outputs : None

In-Outputs :

- `m_workMode` $\mapsto$ m_workMode : $\mathbb{N}$

States :

- `m_countMode` $\mapsto$ m_countMode : $\mathbb{N}$
- `m_countPublic` $\mapsto$ m_countPublic : $\mathbb{N}$

Parameters :

- `time_D2P` $\mapsto$ time_D2P : $\mathbb{N}$
- `time_D2P_overtime` $\mapsto$ time_D2P_overtime : $\mathbb{N}$

2 AG Contract

Assumptions : True

Guarantees :

$$Guarantees \triangleq \exists\, \mathsf{countMode} = \mathsf{m_countMode} + 1 \cdot \big(Case_1 \vee Case_2 \vee Case_3 \vee Other\big)$$

$$Case_1 \triangleq (\mathsf{m_workMode} = 0) \wedge DAMP$$

$$\begin{aligned}
DAMP \triangleq\ & \textsf{if } \max(\mathsf{pGyroRate[0]}, \mathsf{pGyroRate[1]}, \mathsf{pGyroRate[2]}) < 0.15 \textsf{ then} \\
& \quad \textsf{if } \mathsf{m_countPublic} + 1 > \mathsf{time_D2P} \ \vee\ \mathsf{countMode} > \mathsf{time_D2P_overtime} \textsf{ then} \\
& \qquad \mathsf{m_workMode}' = 17 \\
& \qquad \wedge\, \mathsf{m_countMode}' = 0 \\
& \qquad \wedge\, \mathsf{m_countPublic}' = 0 \\
& \quad \textsf{else} \\
& \qquad \mathsf{m_workMode}' = \mathsf{m_workMode} \\
& \qquad \wedge\, \mathsf{m_countMode}' = \mathsf{countMode} \\
& \qquad \wedge\, \mathsf{m_countPublic}' = \mathsf{m_countPublic} + 1 \\
& \quad \textsf{endif} \\
& \textsf{else if } \mathsf{m_countPublic} > \mathsf{time_D2P} \ \vee\ \mathsf{countMode} > \mathsf{time_D2P_overtime} \textsf{ then} \\
& \qquad \mathsf{m_workMode}' = 17 \\
& \qquad \wedge\, \mathsf{m_countMode}' = 0 \\
& \qquad \wedge\, \mathsf{m_countPublic}' = 0 \\
& \quad \textsf{else} \\
& \qquad \mathsf{m_workMode}' = \mathsf{m_workMode} \\
& \qquad \wedge\, \mathsf{m_countMode}' = \mathsf{countMode} \\
& \qquad \wedge\, \mathsf{m_countPublic}' = \mathsf{m_countPublic} \\
& \quad \textsf{endif} \\
& \textsf{endif}
\end{aligned}$$

$$
\begin{aligned}
Case_2 &\triangleq (\text{m_workMode} = 17) \land PITCH \\
PITCH &\triangleq \exists\ \text{workMode_temp}, \text{countMode_temp}, \text{countPublic_temp} \cdot (PITCH_1 \land PITCH_2) \\
PITCH_1 &\triangleq \text{if flgSP} = 235 \text{ then} \\
&\quad \text{if } |\text{piyaw}| > 0.25 \text{ then} \\
&\quad\quad \text{if } (\text{m_countPublic} > 11) \text{ then} \\
&\quad\quad\quad \text{workMode_temp} = 51 \\
&\quad\quad\quad \land\ \text{countMode_temp} = 0 \\
&\quad\quad\quad \land\ \text{countPublic_temp} = 0 \\
&\quad\quad \text{else} \\
&\quad\quad\quad \text{workMode_temp} = \text{m_workMode} \\
&\quad\quad\quad \land\ \text{countMode_temp} = \text{countMode} \\
&\quad\quad\quad \land\ \text{countPublic_temp} = \text{m_countPublic} + 1 \\
&\quad\quad \text{endif} \\
&\quad \text{else} \\
&\quad\quad \text{workMode_temp} = \text{m_workMode} \\
&\quad\quad \land\ \text{countMode_temp} = \text{countMode} \\
&\quad\quad \land\ \text{countPublic_temp} = \text{m_countPublic} \\
&\quad \text{endif} \\
&\text{else} \\
&\quad \text{workMode_temp} = \text{m_workMode} \\
&\quad \land\ \text{countMode_temp} = \text{countMode} \\
&\quad \land\ \text{countPublic_temp} = 0 \\
&\text{endif} \\
PITCH_2 &\triangleq \text{if countMode_temp} > 4500 \text{ then} \\
&\quad \text{m_workMode}' = 34 \\
&\quad \land\ \text{m_countMode}' = 0 \\
&\quad \land\ \text{m_countPublic}' = 0 \\
&\text{else} \\
&\quad \text{m_workMode}' = \text{workMode_temp} \\
&\quad \land\ \text{m_countMode}' = \text{countMode_temp} \\
&\quad \land\ \text{m_countPublic}' = \text{countPublic_temp} \\
&\text{endif}
\end{aligned}
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\begin{aligned}
Case_3 &\triangleq (\mathsf{m_workMode} = 34) \land ROLL \\
ROLL &\triangleq \exists\, \mathsf{workMode_temp}, \mathsf{countMode_temp}, \mathsf{countPublic_temp} \cdot (ROLL_1 \land ROLL_2) \\
ROLL_1 &\triangleq \textbf{if } \mathsf{flgSP} = 235 \textbf{ then} \\
&\quad \textbf{if } |\mathsf{royaw}| > 1 \textbf{ then} \\
&\quad\quad \textbf{if } \mathsf{m_countPublic} > 11 \textbf{ then} \\
&\quad\quad\quad \mathsf{workMode_temp} = 51 \\
&\quad\quad\quad \land\, \mathsf{countMode_temp} = 0 \\
&\quad\quad\quad \land\, \mathsf{countPublic_temp} = 0 \\
&\quad\quad \textbf{else} \\
&\quad\quad\quad \mathsf{workMode_temp} = \mathsf{m_workMode} \\
&\quad\quad\quad \land\, \mathsf{countMode_temp} = \mathsf{countMode} \\
&\quad\quad\quad \land\, \mathsf{countPublic_temp} = \mathsf{m_countPublic} + 1 \\
&\quad\quad \textbf{endif} \\
&\quad \textbf{else} \\
&\quad\quad \mathsf{workMode_temp} = \mathsf{m_workMode} \\
&\quad\quad \land\, \mathsf{countMode_temp} = \mathsf{countMode} \\
&\quad\quad \land\, \mathsf{countPublic_temp} = \mathsf{m_countPublic} \\
&\quad \textbf{endif} \\
&\textbf{else} \\
&\quad \mathsf{workMode_temp} = \mathsf{m_workMode} \\
&\quad \land\, \mathsf{countMode_temp} = \mathsf{countMode} \\
&\quad \land\, \mathsf{countPublic_temp} = 0 \\
&\textbf{endif} \\
ROLL_2 &\triangleq \textbf{if } \mathsf{countMode_temp} > 5000 \textbf{ then} \\
&\quad \mathsf{m_workMode}' = 34 \\
&\quad \land\, \mathsf{m_countMode}' = 0 \\
&\quad \land\, \mathsf{m_countPublic}' = 0 \\
&\textbf{else} \\
&\quad \mathsf{m_workMode}' = \mathsf{workMode_temp} \\
&\quad \land\, \mathsf{m_countMode}' = \mathsf{countMode_temp} \\
&\quad \land\, \mathsf{m_countPublic}' = \mathsf{countPublic_temp} \\
&\textbf{endif}
\end{aligned}
$$

$$
\begin{aligned}
Other \triangleq\ & \mathsf{m_workMode} \neq 0 \\
& \land\, \mathsf{m_workMode} \neq 17 \\
& \land\, \mathsf{m_workMode} \neq 34 \\
& \land\, \mathsf{m_workMode}' = \mathsf{m_workMode} \\
& \land\, \mathsf{m_countMode}' = \mathsf{countMode} \\
& \land\, \mathsf{m_countPublic}' = \mathsf{m_countPublic}
\end{aligned}
$$